
COMMUNITY SORTING ACROSS PLATFORMS

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ABSTRACT

How do political communities move between social media platforms? This paper develops and tests a modified Tiebout model of community sorting across online platforms, conceptualizing platforms as quasi-governments offering public goods in the form of attention distribution to user-generated content. Using agent-based computational simulations, I examine how different platform governance structures—direct voting, coalition formation, and algorithmic recommendation—affect the distribution of utility among communities. The results demonstrate that platform consolidation, extremist presence, and governance mechanisms significantly shape community outcomes. Extremist communities act as “utility vampires,” extracting utility from mainstream communities through hostile action. The simulations reveal that an online informational environment with few platform options, vague content moderation policies, and limited community autonomy drives political communities toward radicalization, while platform diversity and coalition-based governance can mitigate these effects.

Keywords: social media platforms, community sorting, Tiebout model, content moderation, online extremism, agent-based modeling

1 Introduction

Extremist communities do not simply get “deplatformed” and disappear. Instead, many extremist movements appear to cycle across the platform ecosystem: they concentrate on sympathetic peripheral sites, coordinate harassment or recruitment efforts aimed at mainstream audiences, retreat when sanctioned, and then re-enter mainstream spaces when governance conditions change. This pattern is documented by researchers and journalists tracking organized off-site action, including coordinated returns to major platforms for messaging, recruitment, or harassment (Lavin, 2020; Gais and Squire, 2021). The empirical puzzle is not whether online communities move, but why extremist communities move *in this particular way*—and what about the structure of the platform system enables or constrains these strategies (Dehghan and Nagappa, 2022a).

Existing research speaks to pieces of this puzzle, but it does not explain the *system-level* movement patterns that make extremist cycling possible. The deplatforming literature largely treats platform access as a binary condition: actors are either on a mainstream platform or removed from it, after which attention shifts to evasion, reconstitution, or migration to alternative venues (Dehghan and Nagappa, 2022a). Content-moderation and platform-governance research, by contrast, typically focuses on within-platform rulemaking and enforcement, emphasizing the discretionary and contested nature of moderation on general-purpose platforms (Klonick, 2017). Neither framework models how the *mix* of platform types, governance structures, and community compositions creates incentives for extremist groups to concentrate, raid, retreat, and re-enter across multiple venues (Lavin, 2020; Gais and Squire, 2021).

More broadly, political communication work makes clear that where audiences are located matters for politics—platforms can facilitate mobilization and protest, shift public attention, enable surveillance and repression, and amplify or silence marginalized voices (Tufekci, 2017; Barberá et al., 2022; Gohdes, 2020; Pan et al., 2022; Reny and Newman, 2021; Larson et al., 2019; Bonilla and Tillery, 2020; Pan and Siegel, 2020). But these literatures rarely treat the platform ecosystem itself as an object of theory: a competitive environment in which communities endogenously sort across platforms, and in which changes in consolidation versus diversity, permissive versus exclusionary moderation, and governance mechanisms reshape the strategic interaction between extremists and mainstream communities. Explaining extremist cycling therefore requires a model of *inter-platform dynamics*, not only intra-platform policy.

To model community sorting across a system of heterogeneous platforms, I adapt the Tiebout model of citizen sorting across jurisdictions (Tiebout, 1956). Tiebout (1956) showed that if citizens can move freely among municipalities offering different bundles of public goods, they can “vote with their feet,” producing efficient sorting onto preferred jurisdictions. I reconceptualize platforms as quasi-governments offering public goods in the form of attention distribution to user-generated content (UGC), and communities as mobile citizen-voters choosing among these jurisdictions. This theoretical move captures two features missing from existing accounts: communities respond to a *bundle* of governance choices rather than any single policy, and their movement decisions are shaped by the composition of the *entire* platform system, not only the platform they currently occupy.

Within this framework, platform-system structure shapes extremist strategy and, through it, the distribution of attention and safety for mainstream communities. Unlike classic citizen-sorting models, extremists can generate utility by imposing costs on other groups. In the simulations, extremist communities act as “utility vampires,” extracting utility from mainstream communities through hostile action. As a result, some platform ecosystems generate a stable cycle of extremist concentration, raiding, and re-entry, while others dampen it.

My argument is threefold. First, community sorting across platforms is sensitive to governance type and to system concentration: more diverse platform systems produce more efficient sorting, increasing the ability of communities to find jurisdictions that match their preferences. Second, and more importantly, platform-system structure shapes extremist strategy. Extremist communities act as “utility vampires,” extracting value from mainstream communities, and the mix of platform governance types determines whether they concentrate in enclaves, raid mainstream spaces cyclically, or diffuse more evenly across the ecosystem. Third, coalition-based

governance—where communities form movements to shift platform policy—paradoxically provides the strongest protection for mainstream communities by hardening platforms into ideologically coherent spaces that extremists either dominate or abandon.

To demonstrate these dynamics, I develop an agent-based model (ABM) of community sorting across a system of platforms that vary in governance type. In the model, mainstream and extremist communities “vote with their feet” by moving among platforms that allocate attention to user-generated content under one of three governance regimes: direct voting, coalition formation, or algorithmic recommendation. I run three simulation experiments. The first establishes baseline sorting dynamics and asks whether communities efficiently sort onto preferred platforms under each governance regime. The second varies platform-system structure—in particular, concentration versus diversity and the degree of community autonomy—to identify the conditions under which sorting becomes inefficient and radicalization pressures emerge. The third introduces extremists as strategic actors that can extract utility from mainstream communities through hostile action, and tests how different mixes of governance regimes shape whether extremists concentrate in enclaves, cyclically raid mainstream spaces, or diffuse across the ecosystem. The policy-relevant implication is that online environments with few platform options, vague moderation, and limited community autonomy systematically worsen outcomes for mainstream communities while creating structural advantages for extremists.

2 Model

I model community sorting across social media platforms as a modified Tiebout process. Tiebout (1956) argued that if citizens can move freely between municipalities offering different bundles of public goods, they sort themselves efficiently onto jurisdictions matching their preferences. Subsequent refinements have relaxed or extended the original assumptions in various ways (Bewley, 1981; Wooders, 1994; Konishi, 1996). Following this work, I adapt the Tiebout framework to a system of online platforms, reconceptualizing platforms as quasi-governments offering public goods in the form of attention distribution to user-generated content, and communities as mobile agents choosing among these jurisdictions. Two features of the online setting make the Tiebout framework especially apt: movement between platforms is low-cost relative to physical relocation between municipalities, and the “public goods” offered by platforms—content moderation rules, discourse architecture, community autonomy—are experienced collectively by all tenant communities.

The model proceeds in three stages. First, I define the agents (communities and platforms), the structure of the public good space, and the utility functions that govern community behavior (§2.1–2.2). Second, I specify three platform governance mechanisms that determine how public good bundles change in response to community composition (§2.3). Third, I describe the Tiebout cycle: the sequence of governance updates, utility calculations, search, and relocation that constitutes a single simulation step (§2.4). I close with a brief discussion of how each model component maps to observable features of real platforms (§2.5).

2.1 Agents and Environment

Communities. Let $\mathcal{C} = \{c_1, \dots, c_{N_c}\}$ be a set of N_c communities. Each community is a unitary agent with a preference vector $B_c \in \{0, 1\}^{N_a}$, where N_a is the dimensionality of the public good space. Each binary element b_{ca} represents a preference over a specific platform governance decision: whether to amplify or suppress a type of content, enable or disable a community management feature, and so forth. Communities are given and exogenous to the model; I do not model community formation, dissolution, or internal preference change.¹

Communities are one of three types: *mainstream* ($c_m \in \mathcal{C}_m$), *ideologue* ($c_i \in \mathcal{C}_i$), or *griever* ($c_g \in \mathcal{C}_g$), where $\mathcal{C} = \mathcal{C}_m \cup \mathcal{C}_i \cup \mathcal{C}_g$. The proportion of each extremist type is given by ρ_i and ρ_g respectively, with $\rho_m = 1 - \rho_i - \rho_g$. I describe the behavioral differences between these types in §2.2 below.

Mainstream community preferences are drawn uniformly at random from $\{0, 1\}^{N_a}$. Both types of extremist communities have maximally polarized preferences: $B_{c_e} = \mathbf{0}^{N_a}$ or $B_{c_e} = \mathbf{1}^{N_a}$, assigned randomly at initialization. This follows the spatial modeling convention of placing extremists at the boundaries of the preference space, but is supplemented by a richer behavioral model described below.

Platforms. Let $\mathcal{P} = \{p_1, \dots, p_{N_p}\}$ be a set of N_p platforms. Each platform offers a public good bundle $B_p \in \{0, 1\}^{N_a}$, initialized randomly. Platforms are passive: they do not strategically adjust bundles to attract or repel communities.² Each platform has a governance type $\tau_p \in \{D, K, A\}$ —direct voting, coalition formation, or algorithmic recommendation—that determines how B_p changes in response to tenant community composition. These mechanisms are described in §2.3.

Public goods. The bundle B_p represents the combination of content moderation rules, discourse architecture, and community autonomy features offered by a platform. Content moderation rules are the legalistic rules directing platform enforcement processes, ranging from vague standards allowing wide discretion to specific proscriptions of user activity. Discourse architecture refers to designed features of the platform interface that shape user interaction: media type, synchronicity, context collapse, visibility mechanisms. Community autonomy refers to the degree to which users can self-sort, build collective identity, and govern their own boundaries—through built-in community management tools, moderation delegation, or the absence thereof. Each binary variable b_{pa} represents a platform decision along one of these dimensions. Low-dimensional bundles ($N_a \approx 3\text{--}5$) can be interpreted as capturing major moderation categories; high-dimensional bundles ($N_a \geq 20$) capture finer-grained discourse architecture and autonomy decisions.

¹Online communities are, of course, not unitary actors. Membership is fluid, identities are multiple, and intra-community preference aggregation is itself a complex process (Stryker, 2004; Stets and Burke, 2000). I adopt the unitary actor assumption because the model’s focus is platform-community interaction, not intra-community dynamics. This trades precision at the community level for tractability at the system level, a trade-off familiar from structural models in international relations (Waltz, 1997).

²This assumption relocates agency for platform change to communities rather than platform operators. It is a significant simplification; platforms make rapid institutional and procedural changes in response to user behavior, competitive pressures, and political demands (Klonick, 2017). However, the focus of this model is how communities behave given platform structures, not how platforms compete for communities.

2.2 Utility Functions

Base utility. All communities derive utility from the match between their preferences and the public good bundle offered by their current platform. The base utility function is:

$$u_{\text{base}}(c, p) = \sum_{a=1}^{N_a} \delta(b_{pa}, b_{ca}) \quad (1)$$

where δ is the Kronecker delta. This captures the intuition that communities gain utility when platform governance aligns with their preferences: when content they want amplified is amplified, when moderation rules protect rather than suppress their activity, and when discourse architecture serves their communicative needs.

Mainstream communities. When no extremist communities are present, the full utility function for mainstream communities is simply $u_{c_m} = u_{\text{base}}(c_m, p)$. When extremist communities are present, mainstream communities suffer a utility penalty proportional to their exposure to extremists:

$$u_{c_m} = u_{\text{base}}(c_m, p) - \alpha \cdot \frac{N_{c_e}(c_m)}{N_{c_e}(c_m) + N_{c_m}(c_m)} \quad (2)$$

where $N_{c_e}(c_m)$ is the count of extremist neighbors (of either type) and $N_{c_m}(c_m)$ is the count of mainstream neighbors, as defined by the platform’s governance type (see below). The fraction represents the extremist *share* of a community’s neighborhood. The parameter $\alpha > 0$ controls the intensity of the penalty: the maximum utility loss a mainstream community can suffer from extremist exposure, experienced when its neighborhood is entirely extremist.

Extremist communities. I draw on the interdisciplinary literature on contemporary political extremism, which defines extremism as the belief that in-group survival is inseparable from hostile action towards the out-group (Berger, 2018; Gaudette et al., 2021). In this framework, extremism is not merely an ideological position at the boundary of a preference space, but a commitment to a repertoire of action—harassment, hate speech, and other behaviors that degrade the experience of targeted communities. Online extremists act as *utility vampires*, extracting value from the communities they interact with.

I model two types of extremist communities that differ in how they weight policy preferences against parasitic behavior:

Ideologue communities prioritize favorable platform governance. They seek platforms whose public good bundles align with their (maximally polarized) preferences. Their utility function is:

$$u_{c_i} = u_{\text{base}}(c_i, p) + \alpha_i \cdot \frac{N_{c_m}(c_i)}{N_{c_m}(c_i) + N_{c_e}(c_i)} \quad (3)$$

with a low value of α_i , meaning parasitism contributes relatively little to total utility. Ideologues are analogous to communities on platforms like early Gab, which sought a space with favorable

moderation rules and discourse architecture aligned with far-right ideology (Jasser et al., 2023). They benefit from proximity to mainstream communities but primarily seek policy alignment.

Griever communities prioritize hostile interaction with mainstream targets. Their utility function has the same form:

$$u_{c_g} = u_{\text{base}}(c_g, p) + \alpha_g \cdot \frac{N_{c_m}(c_g)}{N_{c_m}(c_g) + N_{c_e}(c_g)} \quad (4)$$

with a high value of α_g , meaning total utility is dominated by the availability of mainstream targets. Grievors are analogous to communities organized around targeted harassment campaigns, such as those originating from Kiwi Farms or organized “raids” from peripheral platforms to mainstream spaces (Lavin, 2020; Gais and Squire, 2021). They care less about platform governance and more about access to victims.

The distinction between ideologues and grievors is controlled by the α parameter. When α is low relative to N_a (the maximum base utility), communities behave as ideologues; when α is high, they behave as grievors. In the simulation, I vary α parametrically across $\{2, 5, N_a\}$ to explore the full range of extremist strategies and the transition between them. These values roughly correspond to settings where vampirism contributes 20%, 50%, or 100% of maximum base utility, respectively.

Utility bounds. Under these functions, utility is bounded for all community types. Mainstream utility falls in $[-\alpha, N_a]$; extremist utility falls in $[0, N_a + \alpha]$. This ensures comparability across platform sizes and configurations, unlike the unbounded linear scaling of previous specifications.

2.2.1 Neighborhoods and Platform Governance

The definition of “neighboring” communities—those who interact and can affect each other’s utility—varies by platform governance type. This is a key theoretical mechanism: different governance structures create different interaction topologies, which in turn determine how extremist parasitism operates and how effectively platforms can insulate mainstream communities from harm.

- **Direct platforms** ($\tau = D$): all communities on the platform are mutual neighbors. The platform is a single undifferentiated public space. Every community is exposed to every other community, and the extremist share of any community’s neighborhood is simply the platform-wide ratio N_{c_e}/N_c . Direct platforms offer no structural protection against extremist exposure.
- **Coalition platforms** ($\tau = K$): communities within the same winning coalition are neighbors. Communities not belonging to any winning coalition are exposed to all platform communities (the unprotected default). If coalition formation effectively segments mainstream and extremist communities into separate movements, mainstream communities within mainstream-dominated coalitions experience zero extremist exposure. The coalition mechanism thus provides a *potential* insulating structure—but only if coalitions form along community-type lines.

- **Algorithmic platforms** ($\tau = A$): communities within the same SVD-assigned group are neighbors. The recommendation algorithm determines who interacts with whom, and communities cannot choose or observe their group assignment. If the algorithm clusters communities by preference similarity, and extremists have distinctive preference profiles, algorithmic grouping may isolate extremists from mainstream communities. However, new arrivals face the “cold start” problem: without sufficient interaction history, the algorithm may assign them to groups with dissimilar communities, temporarily exposing mainstream communities to extremist neighbors or vice versa.

The interaction between neighborhood structure and extremist utility creates governance-specific dynamics. On direct platforms, extremists always have access to mainstream neighbors; the only escape for mainstream communities is relocation. On coalition platforms, mainstream communities can organize into protective coalitions, but extremists can also form coalitions that shift the *status quo* bundle. On algorithmic platforms, the quality of the recommendation system determines whether extremists are effectively quarantined or embedded among mainstream targets.

2.3 Platform Governance Mechanisms

Each platform governance type specifies a procedure for updating the public good bundle B_p at the start of each Tiebout cycle. I describe each mechanism, its key parameters, and the intuition it captures.

2.3.1 Direct Voting ($\tau = D$)

Communities vote on each element of the public good bundle independently. For each policy bit b_{pa} , each tenant community casts a vote for the value matching its own preference b_{ca} . The bit is set to the majority preference; ties are broken by random inversion.

This mechanism has no free parameters beyond the structural parameters N_c and N_a . It resembles community-driven content curation systems such as Reddit’s voting mechanism or the editorial consensus processes of federated platforms like Mastodon.

2.3.2 Coalition Formation ($\tau = K$)

Coalitions are ephemeral movements that propose alternative public good bundles and compete for community support. The mechanism proceeds as follows:

1. **Initialization:** g candidate coalitions are created, each with a randomly generated public good bundle $B_k \in \{0, 1\}^{N_a}$.
2. **Fitness evaluation:** each coalition polls all tenant communities. The fitness of coalition k is the sum of Hamming similarities between B_k and the preference vectors of polled communities.
3. **Mutation:** m randomly selected bits in each coalition bundle are flipped. Communities are re-pollled for fitness. The mutation-poll cycle repeats for μ iterations, capturing the iterative, trial-and-error character of online movement formation as coalitions adjust demands in response to community feedback.

4. **Vote:** communities vote on final coalition bundles. The plurality winner replaces the platform *status quo*.

Parameters: $g \in [2, 10]$ (number of coalitions, varied across iterations), $m = 3$ (mutation rate), $\mu = 10$ (mutation iterations). This mechanism captures movement-driven platform change—coordinated campaigns by communities to shift platform moderation rules, discourse architecture, or autonomy features. Online movements form and dissolve quickly; the mutation process produces coalition bundles that are responsive to community preferences but not perfectly optimized, reflecting the emergent and decentralized character of real platform advocacy.

2.3.3 Algorithmic Recommendation ($\tau = A$)

The platform assigns communities to groups and delivers tailored public good bundles per group, approximating the behavior of recommendation-driven content distribution systems. The mechanism is a simplified implementation of the group-specific SVD recommender system described by Bi et al. (2017).

1. **Candidate generation:** the platform generates s candidate policy bundles and polls all tenant communities for preferences.
2. **Clustering:** communities are partitioned into k groups based on preference similarity, where communities within Hamming distance r of each other are assigned to the same group.
3. **Latent factor computation:** per-group latent factors are computed via truncated singular value decomposition (SVD) on the community-bundle preference matrix.
4. **Bundle assignment:** each group is offered the bundle maximizing predicted group utility based on latent factors.
5. **New arrivals:** when communities join the platform, they are assigned to the nearest group by preference distance and receive that group’s bundle.
6. **Update:** group latent factors are recomputed each cycle as community composition changes.

Parameters: $k \in [3, 20]$ (number of groups), $r \in [1, 5]$ (similarity radius), s (number of candidate bundles). Unlike direct voting and coalition formation, communities on the same algorithmic platform may experience different public good bundles depending on their group assignment. This captures the personalization characteristic of contemporary recommendation systems, where platform governance is experienced differently by different users and communities depending on algorithmically determined groupings that may be opaque to the users themselves.

2.4 The Tiebout Cycle

The simulation proceeds through a sequence of discrete Tiebout cycles. Each cycle $t \rightarrow t + 1$ consists of five steps:

1. **Governance update.** For each platform p , execute the governance mechanism τ_p using the current set of tenant communities to update B_p .

2. **Utility computation.** For each community c on platform p , compute u_c using the appropriate utility function (Equations 2–4).
3. **Search.** Each community c currently on platform p evaluates every other platform $q \neq p$ as a potential destination. Crucially, communities observe only the current public good bundle B_q of each alternative platform. They do *not* observe the composition of communities on other platforms—neither the total count nor the breakdown by type. Communities evaluate potential destinations using only base utility: $u_{\text{base}}(c, q) = \sum_{a=1}^{N_a} \delta(b_{qa}, b_{ca})$. The best candidate is $p^* = \arg \max_{q \neq p} u_{\text{base}}(c, q)$.
4. **Strategy selection.** If $u_{\text{base}}(c, p^*) > u_c$ (the community’s full current utility, including any vampirism or penalty terms), the community selects MOVE to p^* . Otherwise, it selects STAY. Ties among equally good candidate platforms are broken randomly.
5. **Relocation.** All communities selecting MOVE relocate simultaneously.

The simulation terminates when no community selects MOVE (equilibrium) or when the step count reaches t_{max} .

Information and search. The information restriction in Step 3 is a deliberate modeling choice. Communities have complete information about the governance outputs of all platforms—they can observe what public good bundles are on offer—but incomplete information about the social environment on those platforms. An extremist community cannot identify which platforms are “target-rich” before arriving; a mainstream community cannot identify which platforms are “safe” from extremist exposure. Any emergent patterns of strategic movement—such as extremist raiding cycles between platform types—arise from the interaction of utility functions, governance mechanisms, and blind search, not from strategic targeting. This is consistent with the empirical observation that communities migrating between platforms often face significant uncertainty about the social environment they will encounter (Fiesler and Dym, 2020; Chandrasekharan et al., 2017).

Simultaneity. All relocations in Step 5 occur simultaneously. A community’s strategy is based on the state of the system at the start of the cycle, not on the anticipated movements of other communities. This prevents strategic sequencing—where communities wait to see where others move—and ensures that the sorting dynamics are driven by platform governance structures rather than inter-community coordination.

2.5 Mapping to Real Platforms

The three governance types correspond to recognizable structures in the contemporary platform landscape. *Direct voting* platforms resemble community-driven curation systems: Reddit’s up-vote and downvote mechanism, Wikipedia’s editorial consensus, or the community-governed instances of federated platforms like Mastodon, where community members directly influence content visibility and moderation priorities. *Coalition formation* platforms capture the dynamics of movement-driven platform change. Organized advocacy campaigns—such as the coalition of trans, K-pop fan, legal, and human rights communities that successfully pushed Twitter to ban deadnaming, or the counter-campaign by transphobic movements that led to its quiet repeal—reshape platform governance through collective action that forms, succeeds or fails, and dissolves. *Algorithmic recommendation* platforms correspond to the contemporary landscape of

TikTok, YouTube, and algorithmically curated feeds, where platform-assigned groupings determine content exposure and communities experience personalized governance bundles shaped by opaque recommendation systems.

The two extremist types also map to observable community behaviors. *Ideologue* communities resemble movements that have sought platforms with favorable governance structures—such as far-right communities that migrated to Gab for its explicitly permissive content moderation policies (Jasser et al., 2023; Dehghan and Nagappa, 2022b). *Griever* communities resemble those organized primarily around harassment of out-group targets, such as Kiwi Farms or the off-platform organizing for “raids” documented by extremism researchers (Lavin, 2020; Gais and Squire, 2021). In practice, many extremist communities exhibit both behaviors; the α parameter allows the model to explore the full spectrum and identify how the balance between ideological commitment and parasitic behavior shapes sorting outcomes.

3 Experimental Design

I use the model described above to conduct two simulation experiments and a systematic sensitivity analysis. The first experiment establishes baseline sorting dynamics across platform governance types and system structures without extremist communities. The second introduces extremist communities and examines how platform system structure shapes the interaction between extremist and mainstream communities. The sensitivity analysis tests the robustness of key findings to variation in model parameters.

All experiments are implemented in *python* using the *AgentPy* modeling library (Foramitti, 2021) and the Exploratory Modeling and Analysis (EMA) Workbench (Bankes, 1993). Each experimental configuration is run for 200 iterations using independent random seeds. All reported outcomes are means across iterations with 95% confidence intervals unless otherwise noted.³

3.1 Outcome Measures

Each experimental configuration produces four categories of outcome:

Efficiency. Average utility per community, computed system-wide and disaggregated by community type (mainstream, ideologue, griever) and by platform governance type (direct, coalition, algorithmic). Because utility functions differ across community types and are bounded differently ($u_{cm} \in [-\alpha, N_a]$; $u_{ce} \in [0, N_a + \alpha]$), I also report *normalized utility*: each community’s realized utility as a proportion of its theoretical maximum. This allows meaningful cross-type comparisons.

Stability. Average number of community relocations before the simulation reaches equilibrium or $t_{\max}$. High relocation counts indicate that the system is slow to settle, imposing costs on communities that must repeatedly adapt to new platform environments. I also report the *settling time*: the step t^* at which 90% of communities have selected their final platform.

³The parameter space is large: N_c and N_p can take any positive integer value and the dimensionality N_a of the preference space is bounded only by system memory. I constrain the experiments to empirically plausible ranges and report sensitivity to parameter variation in §3.4. Simulation code and replication materials are available at [repository].

Distribution. The final allocation of communities across platform types, reported as counts and proportions. In mixed-governance configurations, the distribution reveals which platform types attract or repel communities and whether particular community types concentrate on particular governance structures.

Dynamics. For experiments involving extremist communities, I track the full relocation history of every community across simulation steps. From these histories, I reconstruct *flow matrices* F_t , where entry $F_t(i, j)$ records the number of communities of each type moving from platform i to platform j at step t . These matrices are used to identify emergent movement patterns, including cyclical “raiding” behavior, enclave formation, and the relationship between platform governance type and extremist movement. I also compute *residence times*—the proportion of simulation steps each community spends on each platform—to identify “home” and “target” platforms for extremist communities.

3.2 Experiment 1: Baseline Sorting Across Platform Systems

Question. How do platform governance type and system concentration affect the efficiency and stability of community sorting in the absence of extremist communities?

Design. This experiment compares sorting outcomes across two dimensions: governance composition (homogeneous vs. mixed) and system size (few vs. many platforms). It has six configurations:

Config	Governance	N_p	N_c	N_a	$t_{\max}$
1a	All direct	9	300	10	100
1b	All coalition	9	300	10	100
1c	All algorithmic	9	300	10	100
1d	Mixed (equal thirds)	9	300	10	100
1e	Mixed (equal thirds)	3	300	10	100
1f	Mixed (equal thirds)	27	300	10	100

Table 1: Experiment 1 configurations. No extremist communities ($\rho_i = \rho_g = 0$).

Configurations 1a–1c isolate the effect of governance type by holding system size constant. Configuration 1d serves as the baseline mixed system at the same scale. Configurations 1e and 1f test the effect of system concentration by varying N_p while holding N_c fixed, so that the average number of communities per platform changes (100, 33, and 11 communities per platform respectively). Holding N_c constant across 1d–1f ensures that differences in outcomes are attributable to system structure rather than community density.

Expectations. Consistent with the Tiebout hypothesis, I expect that configurations with more platform options (1f) will produce higher average utility and faster settling than concentrated configurations (1e), as communities have more jurisdictions to sort across. Among governance types, I expect algorithmic platforms to deliver higher utility in multi-platform settings due to personalized bundle delivery, but with greater instability early in the simulation as recommendation systems calibrate. Coalition platforms should exhibit the highest variance in outcomes due to the stochastic character of movement formation.

3.3 Experiment 2: Extremist Dynamics in Mixed Platform Systems

Question. How does platform system structure shape the strategic behavior of extremist communities and the welfare of mainstream communities?

Design. This experiment introduces extremist communities into mixed-governance platform systems and varies three factors: system concentration (N_p), extremist prevalence (ρ_e), and parasitism intensity (α). The experiment uses a factorial design across these three dimensions.

I operationalize the griever/ideologue distinction through variation in α rather than as discrete community types. When α is low ($\alpha = 2$), all extremists behave as ideologues: parasitism contributes at most 20% of maximum base utility, so sorting is dominated by policy preferences. When α is high ($\alpha = N_a = 10$), all extremists behave as grievers: parasitism can contribute as much as base utility itself, so the availability of mainstream targets becomes the primary driver of extremist behavior. The intermediate value ($\alpha = 5$) represents communities that balance ideological commitment with parasitic behavior.

Factor		Levels		
System concentration (N_p)	3	9	27	
Extremist prevalence (ρ_e)	0.05	0.10	0.15	
Parasitism intensity (α)	2	5	10	
<i>Fixed: $N_c = 900$, $N_a = 10$, $t_{\max} = 100$, mixed governance (equal thirds)</i>				

Table 2: Experiment 2 factorial design. $3 \times 3 \times 3 = 27$ configurations, 200 iterations each.

Several design choices require justification. First, I fix $N_c = 900$ across all configurations to ensure that per-platform community counts are large enough for the proportional utility functions to produce stable ratios, even in the $N_p = 27$ condition (where the average platform has ≈ 33 communities). Second, I use mixed-governance systems (equal thirds direct, coalition, algorithmic) in all configurations to isolate the effect of system structure and extremist parameters from governance composition. Third, I set $t_{\max} = 100$ to allow sufficient time for sorting dynamics to develop; results from Experiment 1 will confirm whether this is adequate.

Analysis strategy. This experiment addresses several distinct questions, each requiring a different analytic approach:

How does parasitism intensity interact with system concentration? For each α level, I compare mainstream utility across N_p levels. The key comparison is whether the welfare cost of platform consolidation (fewer platforms) is amplified when extremists are more parasitic. If so, this implies that platform diversification is especially important in media environments where griever-type extremists are prevalent.

Which governance types protect mainstream communities? I disaggregate utility outcomes by platform governance type within each configuration. The current manuscript found that coalition platforms paradoxically protect mainstream communities once extremists are introduced. I test whether this finding holds under the revised utility function and whether it depends on α : coalition platforms may protect against ideologues (who seek policy alignment) differently than against grievers (who seek targets).

Do extremist movement patterns differ by type? Using the flow matrices and residence time measures described in §3.1, I characterize extremist relocation behavior across configurations. Specifically, I test for the “raiding cycle” identified in the earlier version of this analysis—a pattern where extremists concentrate on peripheral direct platforms and periodically move to larger algorithmic platforms to parasitize mainstream communities. The key prediction is that this cycle should be more pronounced at high α (griever behavior) than at low α (ideologue behavior), because grievors have stronger incentives to seek mainstream-heavy platforms regardless of governance bundle.

To formalize the raiding cycle, I compute the autocorrelation of extremist outflow from direct platforms across simulation steps. Significant periodicity (peaks in the autocorrelation function at regular lag intervals) indicates cyclical raiding behavior. I also measure cycle amplitude (the number of extremists relocating per cycle) and test whether amplitude increases over the course of the simulation, which would indicate an escalation dynamic.

Do coalition platforms form enclaves? I examine the final composition of winning coalitions on coalition platforms. If coalitions segment by community type—mainstream-dominated coalitions excluding extremists, and vice versa—this constitutes enclave formation. I measure this using the *type homogeneity* of winning coalitions: the proportion of coalition members that share the same community type (mainstream or extremist). High homogeneity indicates enclave formation; low homogeneity indicates integration.

3.4 Sensitivity Analysis

The model contains several parameters whose values are chosen for tractability or to match rough empirical intuitions rather than estimated from data. To assess the robustness of key findings, I conduct a systematic sensitivity analysis on the baseline Experiment 2 configuration ($N_p = 9$, $\rho_e = 0.10$, $\alpha = 5$).

Parameters varied. I conduct one-at-a-time (OAT) sensitivity analysis on six parameters, varying each across its plausible range while holding all others at their baseline values:

Parameter	Symbol	Baseline	Range
Preference dimensionality	N_a	10	{5, 10, 20, 50}
Number of coalitions	g	5	{2, 5, 10}
Coalition mutation rate	m	3	{1, 3, 5}
SVD groups	k	10	{3, 5, 10, 20}
Similarity radius	r	3	{1, 3, 5}
Maximum simulation steps	$t_{\max}$	100	{50, 100, 200}

Table 3: Sensitivity analysis parameters.

Interaction effects. In addition to OAT analysis, I compute second-order Sobol sensitivity indices for two theoretically important parameter interactions:

1. $\alpha \times N_p$: Does the effect of parasitism intensity depend on system concentration? If so, the welfare consequences of platform consolidation are contingent on the type of

extremist communities present—a finding with direct policy implications for platform regulation.

2. $\alpha \times N_a$: Does the effect of parasitism intensity depend on the dimensionality of the public good space? Higher dimensionality means base utility has greater variance, which changes the relative importance of the parasitism term. If α 's effect diminishes as N_a grows, this would suggest that the griever/ideologue distinction matters most in low-complexity governance environments.

Reporting. For OAT analysis, I report the percentage change in each outcome measure (mainstream utility, extremist utility, average moves, settling time) relative to the baseline when each parameter is set to its minimum and maximum values. For Sobol indices, I report first-order and total-effect indices with bootstrap confidence intervals following ?. Results that are qualitatively sensitive to parameter choices—where the direction of a key finding reverses—are flagged and discussed in the results section.

4 Results

4.1 Experiment 1: Sorting Without Extremists

4.1.1 Result 1: Governance type shapes sorting efficiency, but the ranking depends on system composition

4.1.2 Result 2: Platform diversification improves welfare

4.2 Experiment 2: Extremist Dynamics in Mixed Platform Systems

4.2.1 Result 3: Diversification protects mainstream communities, and the protection premium grows with extremist threat

4.2.2 Result 4: Parasitism intensity produces qualitatively distinct extremist regimes

4.2.3 Result 5: Coalition governance creates protective enclaves

4.2.4 Result 6: Extremists concentrate on direct platforms and exploit the direct-algorithmic interface

4.3 Raiding Dynamics

4.3.1 Result 7: Extremist movement follows a bursty raiding pattern

4.4 Summary

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